Competitive Markets and Market Interventions

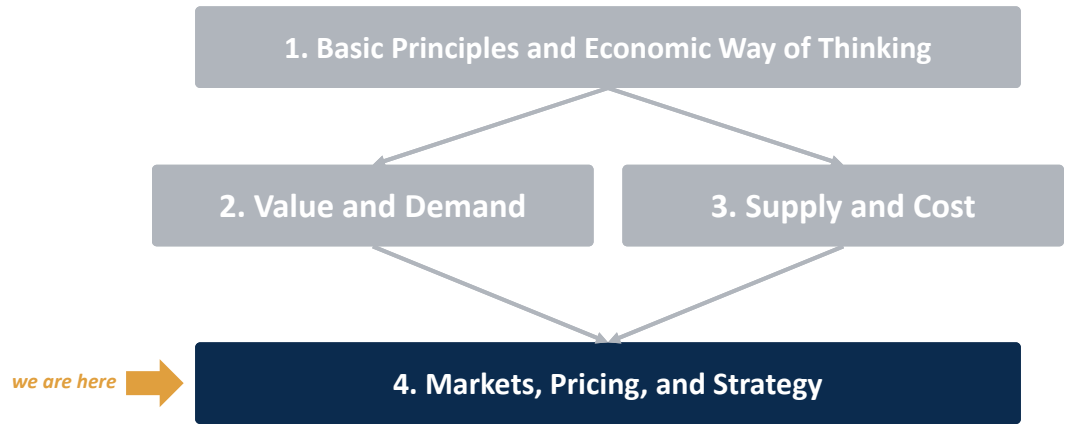
Module 4

Management 405

Prof. Nico Voigtländer · UCLA Anderson

1

Agenda for the Class



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Outline of Module 4

1. **Introduction to Market Structures** — Video 1
   Where a firm sits between price taker and price setter
2. **Perfect Competition** — Video 2
   Many small sellers, an identical product, and free entry
   - 2a **Profit Maximization of a Price Taker in the Short Run** — Video 3
     How much to produce, and when to stop production
   - 2b **Firm-Level and Market Supply** — Video 4
     Marginal cost is one firm's supply curve; market supply sums across firms
   - 2c **Long-Run Competitive Equilibrium** — Video 5
     Entry and exit drive economic profits to zero
3. **Market Distortions and Regulations** — In Class
   Who wins, who loses, and what is lost outright
4. **Externalities** — In Class
   Costs and benefits that land on people outside the deal

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Module 4
In-Class Part

4

Some Applications of the Material Covered in the Videos

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The Four Basic Market Structures

	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of firms	Many	Many	Few	One
Type of products sold	Identical	Differentiated	Identical or Differentiated	"Unique"
Barriers to entry	None	None	Some	Many

More firms, more similar products, easier entry, more competition

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Extent of Market Power: Price Takers vs. Price Searchers

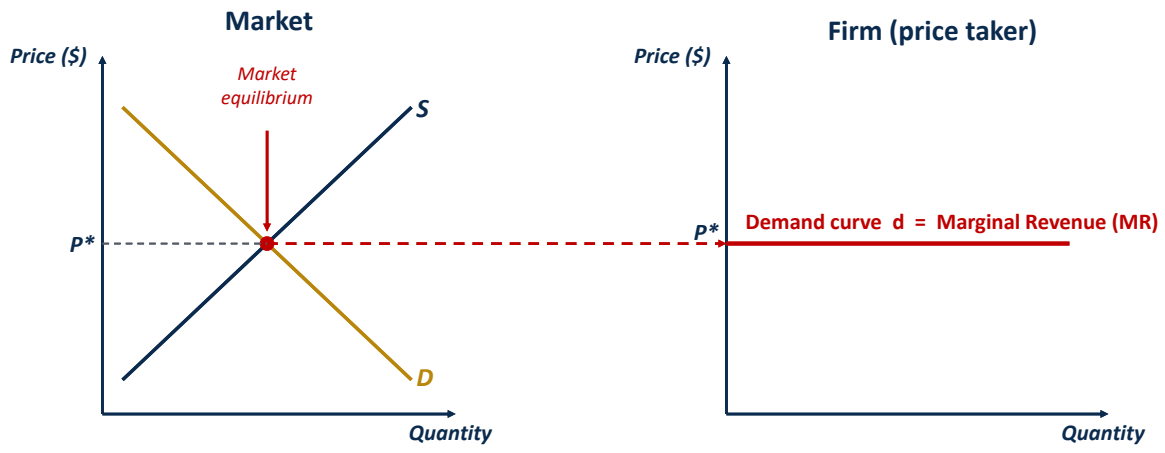
Least Market Power ⟷ **Most Market Power**

Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
(Price Takers)	(Price Searchers)		
		BOEING / AIRBUS	LA DWP
Commodity markets	*Restaurants in WeHo*	*Wide-body aircraft*	*Municipal utilities*

TODAY

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Perfect Competition: The Market and the Firm



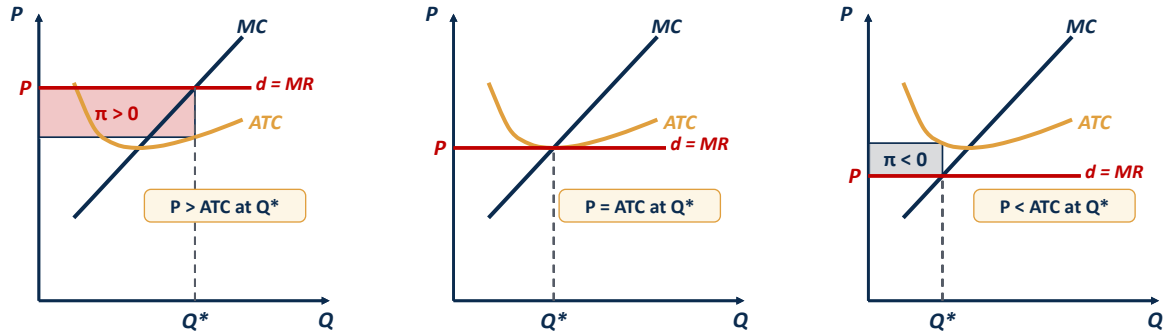
8

Module 4 · Video 3 · Profit Maximization of a Price Taker in the Short Run

Maximizing Profits in the Short Run: The General Case

Profit Levels for Different Market Prices

Positive Profit	Zero Profit	Negative Profit (Loss)

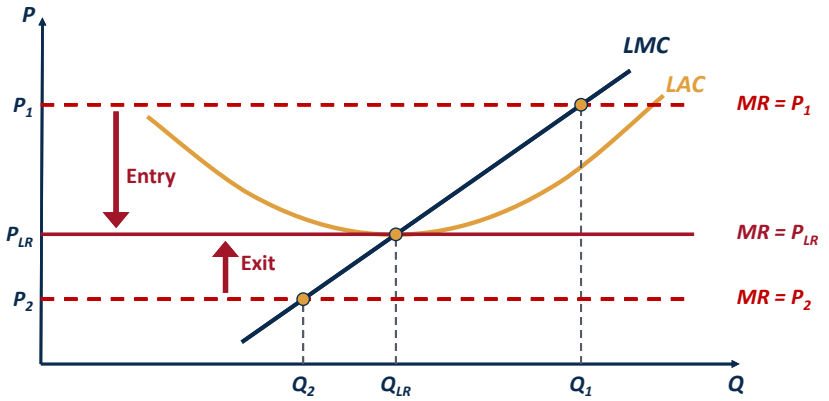


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Module 4 · Video 5 · Long-Run Competitive Equilibrium

Long-Run Competitive Equilibrium With Many Identical Producers

Competitive Markets: Large number of (actual and potential) firms that have the same cost structure and compete with each other



At Q_{LR}: P = LAC = LMC, and economic profit is zero

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Long-Run Equilibrium: The Drug Market

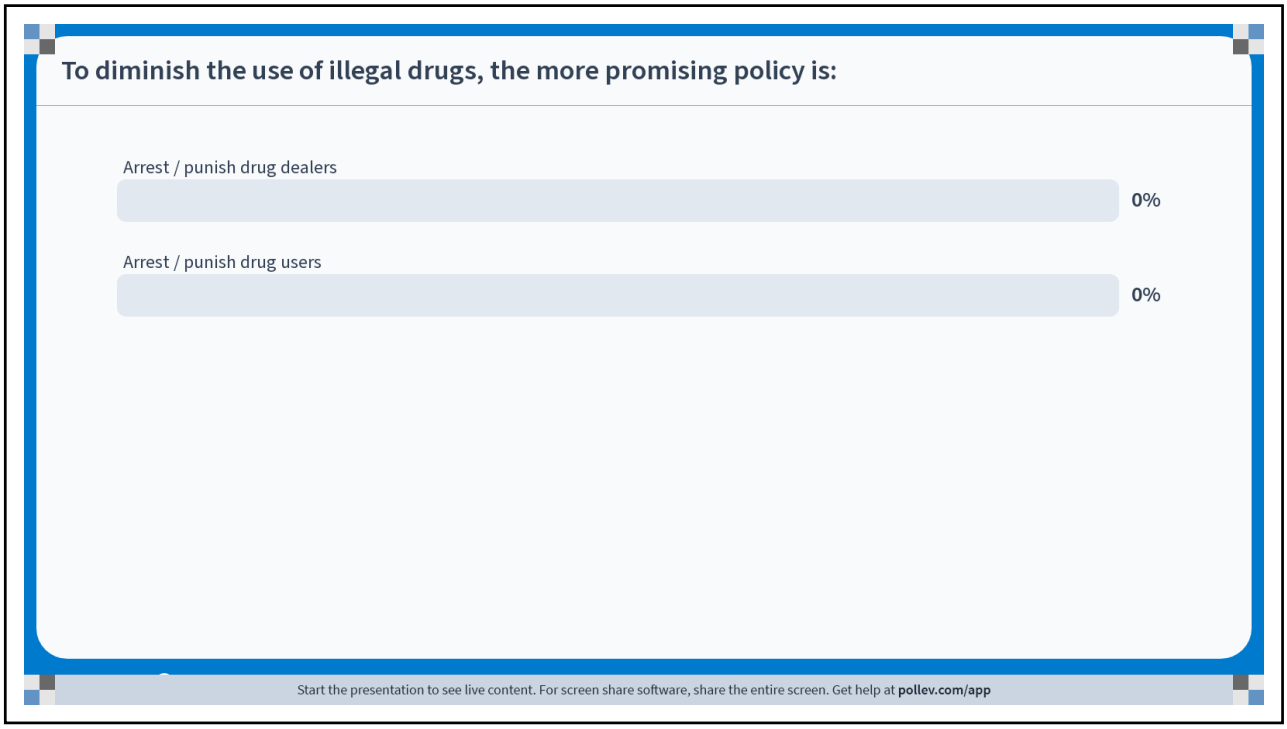
- Can we use the toolbox on the market for illegal drugs?
  - Assume individual dealers are price takers

The government has two one-time options:

1. Arrest the drug dealers
2. Arrest or punish the drug users

Which policy does more to reduce the use of illegal drugs?

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Arrest Drug Dealers

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Arrest / Punish Drug Users

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Outline of Module 4

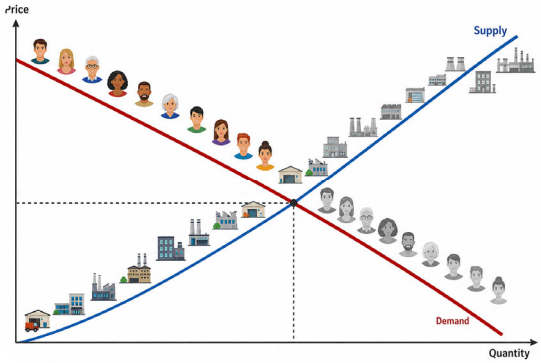
1	Introduction to Market Structures	Video 1
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3	**Market Distortions and Regulations** Who wins, who loses, and what is lost outright	**In Class**
4	Externalities	In Class

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Welfare in a Market Economy and Distortions: Introduction

- In a competitive market, equilibrium is where supply meets demand
- Outside forces can push the market away from it
  - Price controls — rent control, minimum wages
  - Taxes and subsidies
- How do we tell who gains and who loses?
  - Consumer and producer surplus
  - Deadweight loss

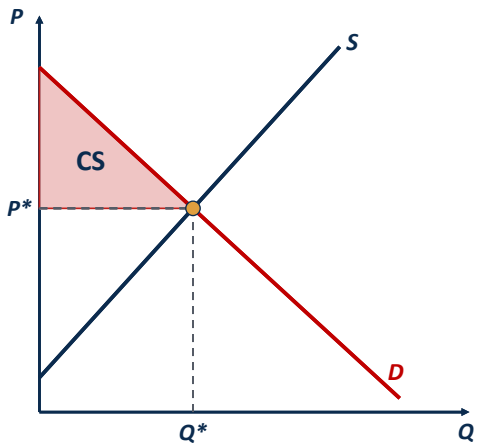
A competitive market with many consumers and producers



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Consumer Surplus

CS: the gap between what consumers would pay (the D curve) and what they do pay (P*)



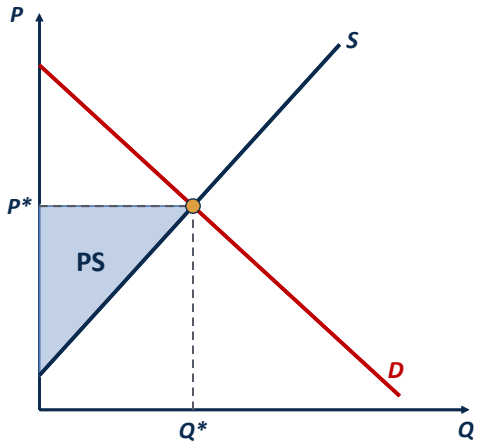
Consumer surplus (CS) is the area below the demand curve and above the price line

What buyers were willing to pay, less what they actually paid

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Producer Surplus

PS: the gap between the price at which producers would sell (the S curve) and what they do receive (P*)



Producer surplus (PS) is the area above the supply curve and below the price line

What sellers received, less the least they would have accepted

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Market Distortions: Deadweight Loss

- The inefficiency that a market intervention creates
- Mutually beneficial transactions that no longer happen
  - Workers willing to work at the market wage and firms willing to hire them — but a minimum wage stands in the way

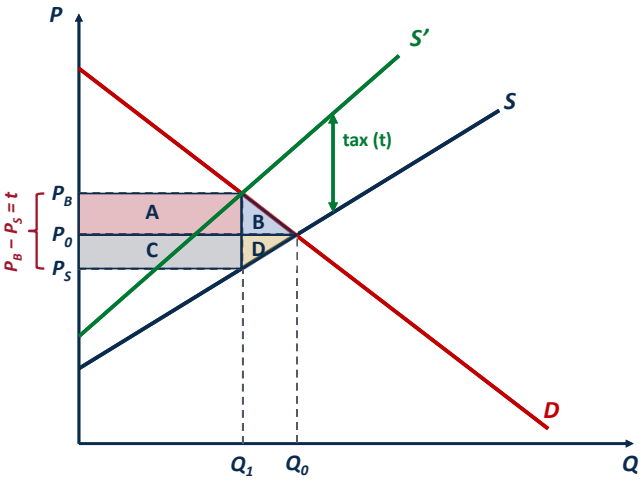


$$\text{Deadweight loss} = \text{Welfare}_{\text{free market}} - \text{Welfare}_{\text{regulation}}$$

$$\text{with} \quad \text{Welfare} = \text{CS} + \text{PS}$$

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Sales Tax: a 10 % Tax on Top of the Listed Price



- *Price received by the seller (P_S) drops.*
- *Total price paid by the buyer (P_B, inclusive of tax) increases, but by less than the tax.*

The buyers lose A + B

The sellers lose C + D

Deadweight loss: B + D

The government gets A + C

Taxes on some goods (e.g. cigarettes, gasoline…) can still make sense due to externalities (more below)

Problem Set 3

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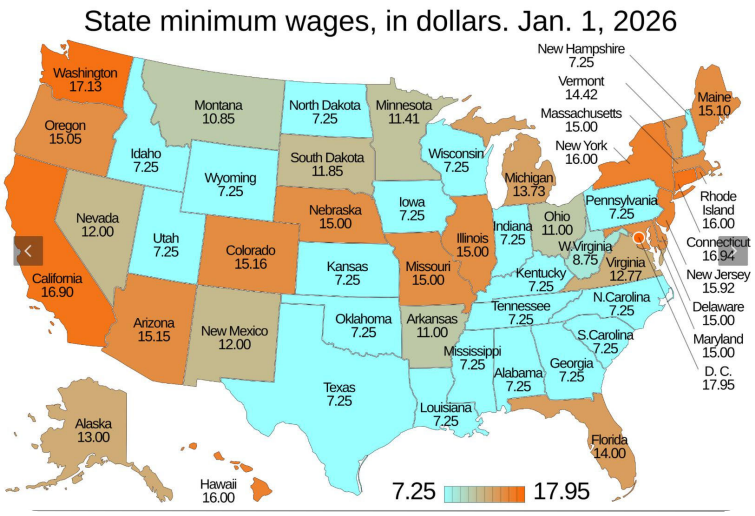
Tax Incidence: Who Really Pays?

Tax incidence: which side of the market — buyers or sellers — ends up bearing the burden of the tax

- Who hands the money to the government does not matter
  - The incidence is the same whether the tax is collected from the buyer or from the seller
- What matters is the elasticity of supply and demand
  - The side that responds less to price bears more of the tax

The less elastic side of the market pays the larger share of a tax

The Minimum Wage: A Popular Policy Tool



Source: Wikipedia / US Department of Labor

Podcast: What Does the Evidence Say on the Minimum Wage?

- What is the expected effect of raising the minimum wage on employment?
  - According to economic theory: Negative
- Evidence: Mixed results
  - Employment does not seem to decline, BUT:
  - Hours worked decline (Seattle study)
    - Experienced workers: moderate decline in hours, so total income ↑
    - Low-skilled, inexperienced workers: substantial decline in hours and thus income ↓
    - New workers: harder to find first job
- What are possible long-run effects of a (high) minimum wage?



SundayReview | OPINION

The Minimum Wage: How Much Is Too Much?

By ALAN B. KRUEGER OCT. 9, 2015

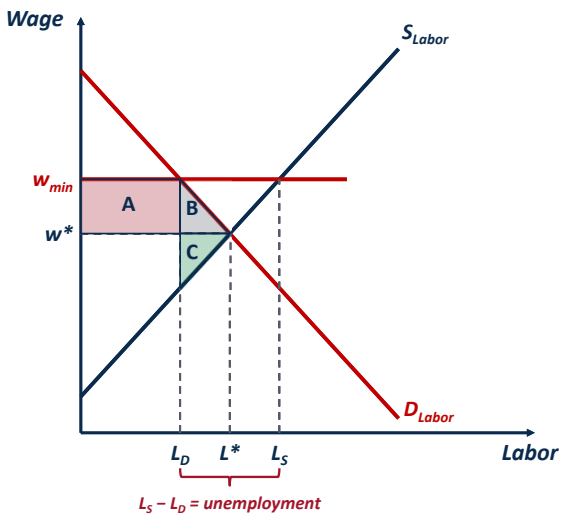
The controversial study showing high minimum wages kill jobs, explained

Fight for $15 advocates are worried.

By Jeff Guo | @_jeffguo | Jun 27, 2017, 12:00pm EDT

One way firms respond to a higher wage floor

Price Floor: The Minimum Wage



Welfare effects

- Some workers win A — they are paid more
- Some workers lose C — they are not hired
- Firms lose A + B
- Deadweight loss: B + C

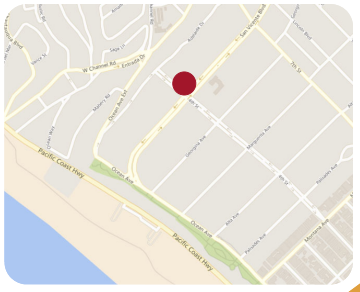
Circumvention of minimum wages?

- Underground labor market
- Cutting non-wage benefits
- Unpaid internships

Guess the Rent!



- 1 bed, 1 bath
- 900 sqft
- 401 San Vicente Blvd., Santa Monica
  - 5-minute walk to the beach
  - Best public school district in West LA



Poll Break

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Guess what's the monthly rent...

(A) >5,000 0%

(B) 3,000-5,000 0%

(C) 1,500-2,999 0%

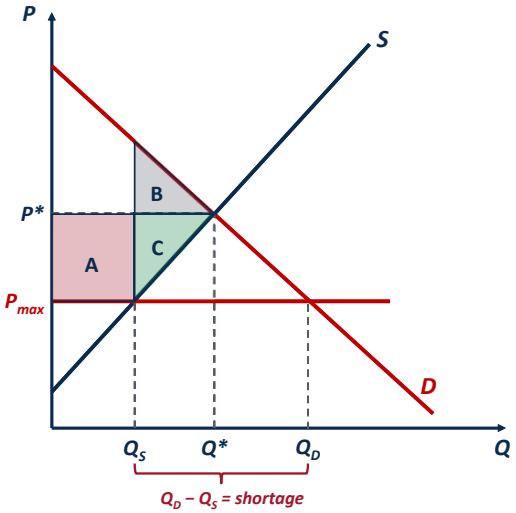
(D) 1,000-1,499 0%

(E) <1,000 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

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Price Ceiling: Rent Control



Welfare effects

- Some renters win A — those who keep a place
- Some renters lose B — they cannot find one
- Landlords lose A + C
- Deadweight loss: B + C

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How Do Landlords React to Rent Control?

- Non-price adjustments (e.g. neglect maintenance)
- Buy the tenants out (legal)
- Harass the tenants (illegal)

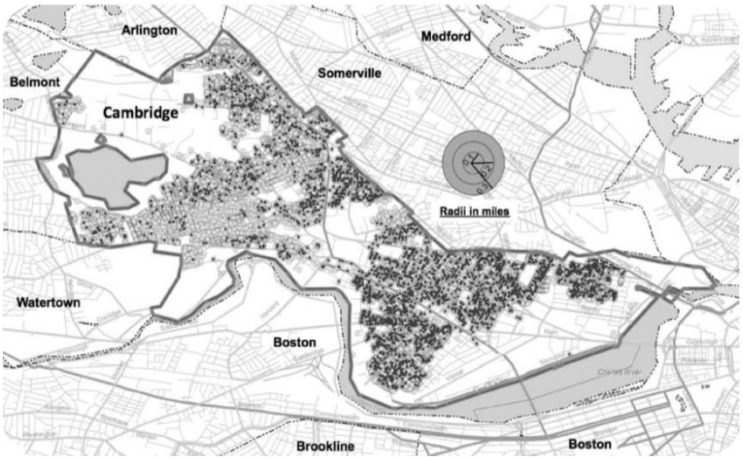




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Evidence on Rent Control: Cambridge, MA

A natural experiment: rent control in Cambridge ended suddenly in 1995



Source: Autor, Palmer and Pathak (2014)

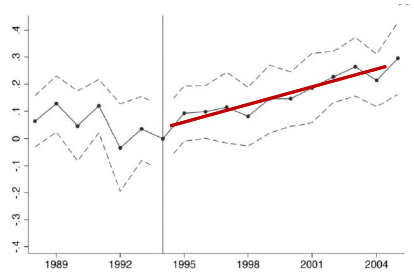
What Happened When Rent Control Ended?

- "Natural" experiment: sudden end of rent control in Cambridge, MA in 1995
- What do you think happened to:
  - Rental price for previously rent-controlled units?
  - Investment in previously rent-controlled units?
  - Price/investment for nearby never-controlled units?

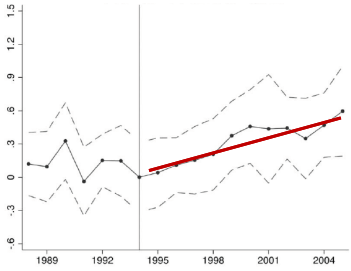
Discussion Break

Evidence on Rent Control: Results

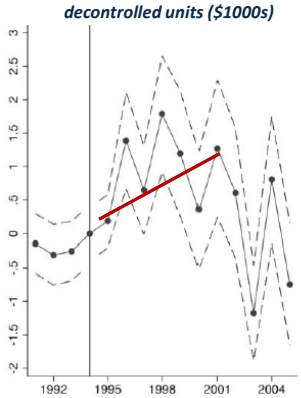
Impact on the price of decontrolled units



Impact on the price of never-controlled units



Impact on investment activities at decontrolled units ($1000s)



Rents rose, and so did investment — including at the units next door

Argentina Scrapped Its Rent Controls



- Argentina repealed its rent-control law at the end of 2023
- Reported since: many more flats listed, and rents lower in real terms
- Not everyone gains — sitting tenants lost their protection

Module 4 · Agenda

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   Costs and benefits that land on people outside the deal

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Module 4 · Externalities

Externalities: Definitions

Externality: A cost or a benefit that falls on someone who is not part of the transaction

- **Negative externality** — A **cost** imposed on an outsider
  - Air pollution from coal-fired power plants
  - The external cost of gasoline is about $1.50 per gallon
- **Positive externality** — A **benefit** conferred on an outsider
  - Your neighbor's well-kept front yard is a pleasure for you to look at too

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Correcting for Externalities

The aim of the intervention: move the market toward the quantity that would be chosen if the outside cost or benefit were counted

Price mechanisms: taxes and subsidies

- Tax the activity that imposes the cost, subsidize the one that confers the benefit

Quantity mechanisms: quotas and mandates

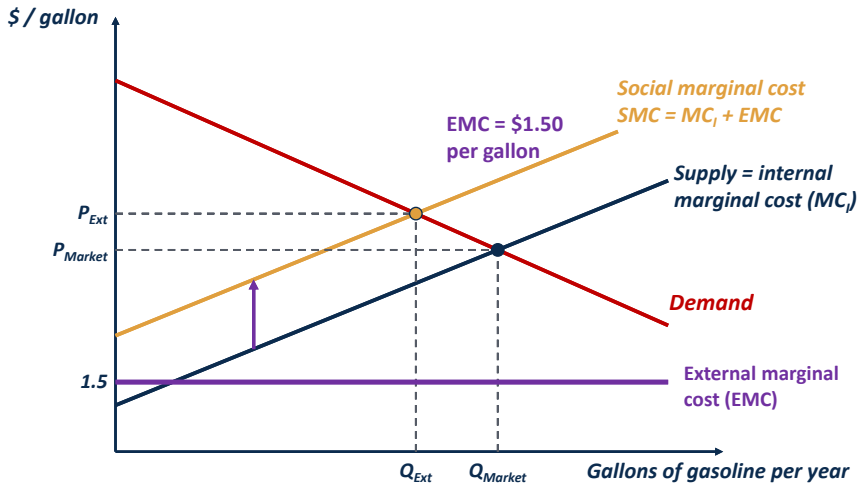
- **Negative externality — cap it:** a limit on the tonnes of sulphur dioxide a power plant may emit in a year
- **Positive externality — require it:** childhood vaccination as a condition of school entry


The New York Times Magazine

IT'S THE ECONOMY

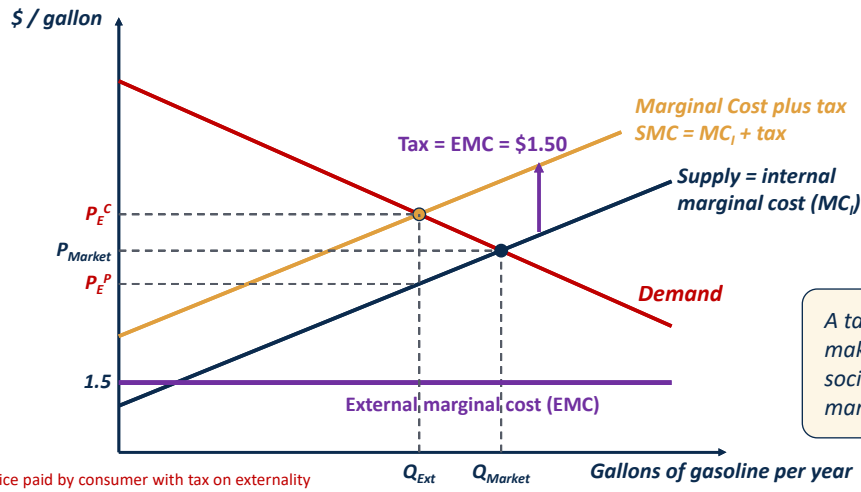
Should We Tax People for Being Annoying?

The Externality: Gasoline



Drivers weigh only their own cost, so the market settles at Q_{Market}. Counting the cost borne by everyone else, the efficient quantity is the smaller Q_{Ext}.

A Carbon Tax to Correct the Negative Externality





Arthur Pigou

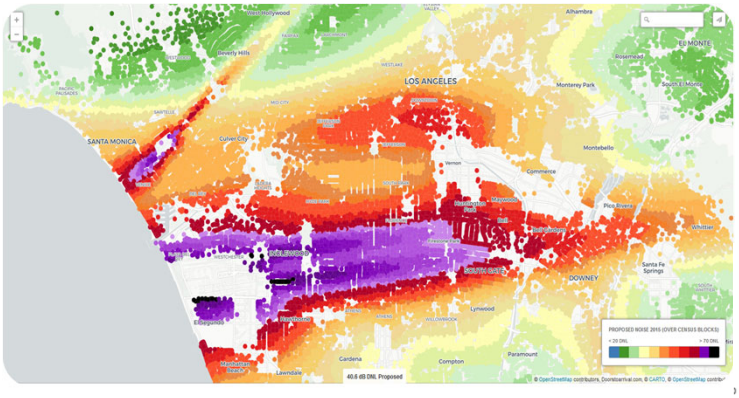
A tax equal to the external cost makes each driver face the social marginal cost, and the market lands on Q_{Ext} on its own.

P_E^C : Price paid by consumer with tax on externality

P_E^P : Price received by producer with tax on externality

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Externality: Airplane Noise



Noise contours around Santa Monica Airport

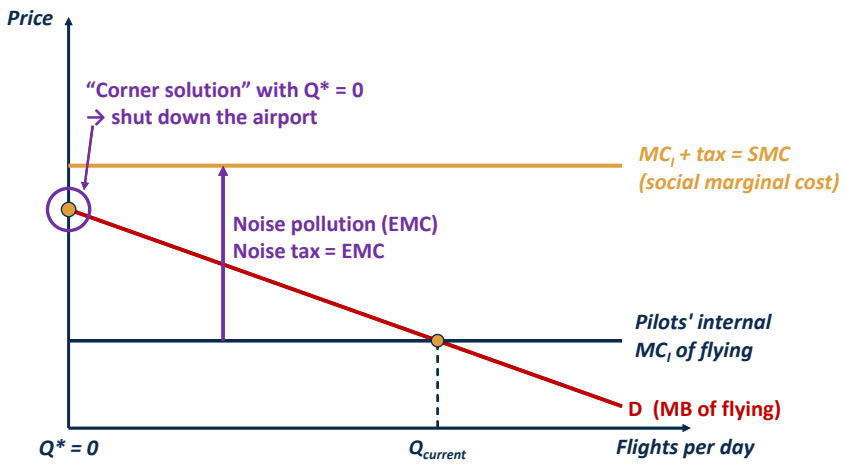
Santa Monica
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Jet Traffic Carries Noise Violations at Santa Monica Airport to New Heights

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A Pigouvian Tax for Noise? The Case of Santa Monica Airport



SMC lies above willingness to pay at every quantity — a corner solution at Q* = 0, so close the airport